

Chloramphenicol (Compound **1**) is an antibiotic that acts against gram-positive and gram-negative bacteria by blocking peptidyl transfer, which inhibits protein synthesis by bacterial ribosomes. However, it tastes bitter, so most patients cannot tolerate it. As a result, synthesis of hydroxyl-substituted chloramphenicol analogues has become of interest because ester chloramphenicol analogues taste better than chloramphenicol itself. Substituents on the primary hydroxyl group of chloramphenicol analogues are typically hydrolyzed quickly *in vivo* to regenerate Compound **1**. Compound **3**, a chloramphenicol analogue, was synthesized by the transesterification of Compound **1** and vinyl propionate (Compound **2**), catalyzed by the enzyme Lip_{BA} lipase (Figure 1).

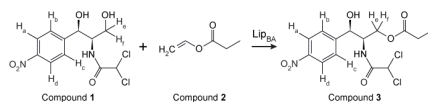


Figure 1 Transesterification of chloramphenicol catalyzed by Lip_{BA}

Although enzyme catalysis allows for efficient regiospecific and stereospecific bond formation, the reaction conditions for the conversion efficiency of Compound **1** to Compound **3**, as well as the purity of Compound **3**, must also be optimized. The reaction solvent, temperature, time, and enzyme concentration were studied to find the conditions that gave the highest conversion rate and purity, with each parameter altered individually while all others were kept constant. The results of this study are shown in Table 1. Reactions in which no enzyme was used yielded no product and are not shown in the table. The purity of Compound **3** was found to be over 90% when 1,4-dioxane was used and over 70% when dichloromethane was used.

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The transesterification reaction was monitored by thin-layer chromatography (Figure 2), and the product was confirmed by proton nuclear magnetic resonance spectroscopy (¹H NMR). The ¹H NMR spectra of Compounds **1** and **3** can be distinguished by the shift of H_e and H_f due to the addition of the electronegative ester functional group.

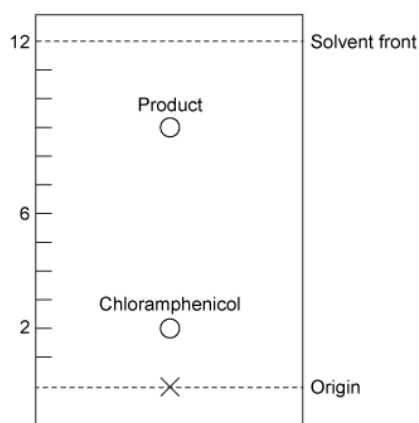


Figure 2 Thin-layer chromatography of the reaction mixture

Based on the thin-layer chromatography plate in Figure 2, which of the following expressions gives the R_f value corresponding to the transesterification product?

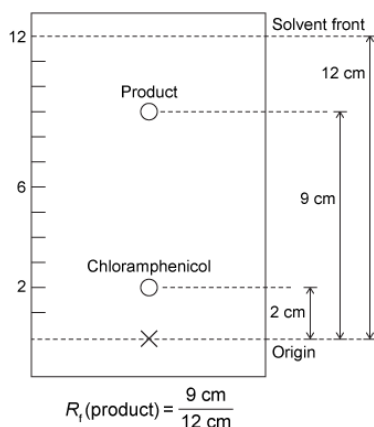
- ☐ A. 2 cm/12 cm [2%]
✔ ☒ B. 9 cm/12 cm [91%]
☐ C. 12 cm/9 cm [4%]
☐ D. 12 cm/2 cm

Correct

91% answered correctly

Explanation:

$$R_f = \frac{\text{distance compound traveled from origin}}{\text{distance from origin to solvent front}}$$



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Thin-layer chromatography (TLC) is a technique used to **separate** compounds based on **polarity**. The mobile phase travels up the stationary phase via **capillary action**. In normal-phase TLC, the stationary phase is made of a polar adsorbent material, typically silica (SiO_2), and the mobile phase is a nonpolar organic solvent. The components of a mixture in the solvent travel up the plate at different rates based on their **polarity**. Nonpolar compounds have less affinity for the polar stationary phase than polar compounds, so they interact with the stationary phase less and travel farther up the plate.

The **R_f value** is expressed by the **ratio** of the distance traveled by the compound of interest to the distance traveled by the mobile phase (solvent front). The R_f value is always **less than 1** and when two compounds are compared, a **larger R_f value** corresponds to the **less polar**, more mobile compound.

Figure 2 shows that the product of transesterification traveled 9 cm up the plate and the solvent traveled 12 cm. Therefore, the R_f value of the product is calculated as the ratio 9 cm/12 cm.

(Choice A) The expression 2 cm/12 cm gives the R_f value of chloramphenicol (starting material) rather than the R_f value of the product.

(Choices C and D) The expressions 12 cm/9 cm and 12 cm/2 cm are both greater than 1, and R_f values must be less than 1. These values are obtained if the ratio in the R_f calculation is reversed (distance solvent traveled to distance compound traveled) for the product and for chloramphenicol, respectively.

Educational objective:

Thin-layer chromatography is a technique used to separate compounds based on polarity. The R_f value is expressed as the ratio of the distance the compound of interest traveled to the distance the mobile phase (solvent front) traveled; its value is always less than 1. A larger R_f value corresponds to a less polar compound, and vice versa.

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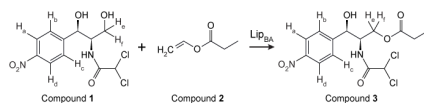


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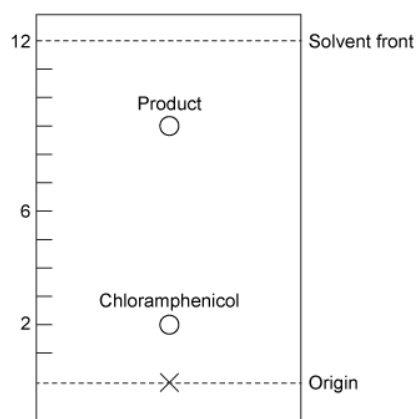


Figure 2 Thin-layer chromatography of the reaction mixture

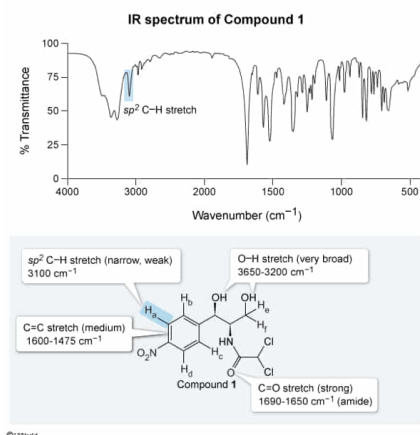
Based on the structure of Compound **1** provided in Figure 1, which stretch gives a narrow absorption of weak intensity at approximately 3100 cm^{-1} in the infrared spectrum?

- ☐ A. The C=C stretch [13%]
☐ B. The C=O stretch [14%]
☒ C. The C-H_a stretch [40%]
☐ D. The O-H stretch [30%]

Correct

40% answered correctly

Explanation:



In infrared (IR) spectroscopy, a sample is irradiated with IR light, and a spectrometer detects and records the percentage of radiation that passes through the sample over a range of frequencies. Different **functional groups absorb IR**

radiation at different **intensities** and **different frequencies**, and therefore the signal from each group appears in a particular region of the **IR spectrum**. The absorption frequency for a particular **functional group** depends on the types of bonds present. Characteristic functional group absorptions include:

- 3650–3200 cm^{-1} (O–H stretch)
- 3300 cm^{-1} (*sp* C–H stretch)
- 3100 cm^{-1} (*sp*² C–H stretch)
- 3000 cm^{-1} (*sp*³ C–H stretch)
- 1810–1650 cm^{-1} (C=O stretch)

Compound **1** has a variety of functional groups, including an aromatic ring, hydroxyl groups, an amide, and an alkyl group. The ***sp*² C–H_a stretch** from an aromatic ring absorbs light at approximately **3100 cm^{-1}** (within the IR spectrum), resulting in a narrow band of weak intensity.

(Choices A, B, and D) C=C (aromatic ring), C=O (amide), and O–H stretches absorb in the 1600–1475 cm^{-1} , 1690–1650 cm^{-1} , and 3650–3200 cm^{-1} regions of the IR spectrum, respectively. The C=C stretch is weak in intensity, the C=O stretch is strong in intensity, and the O–H stretch is broad.

Educational objective:

Functional groups absorb at different intensities and different frequencies in the infrared spectrum, depending on the type of bond present in the particular functional group. Characteristic functional group absorption ranges include 3650–3200 cm^{-1} (O–H stretch), 3300 cm^{-1} (*sp* C–H stretch), 3100 cm^{-1} (*sp*² C–H stretch), 3000 cm^{-1} (*sp*³ C–H stretch), 1810–1650 cm^{-1} (C=O stretch), and 1600–1475 cm^{-1} (aromatic ring C=C stretch).

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 402225

System:

Separation Techniques, Spectroscopy, and Analytical Methods

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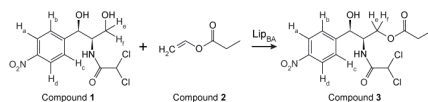


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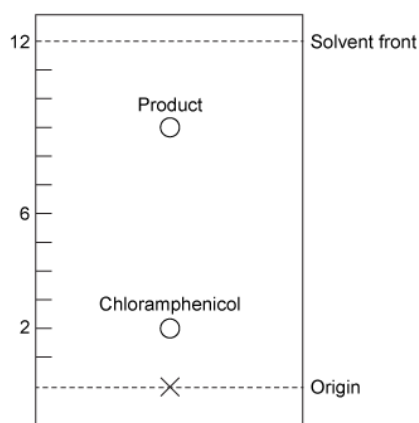


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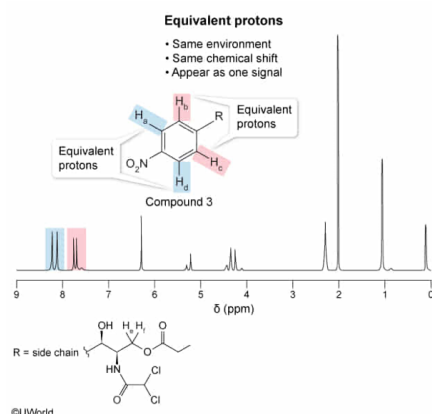
What is the number of signals corresponding to aromatic hydrogen atoms present in the ^1H NMR spectrum of Compound **3**?

- ☐ A. 1 [5%]
✔ ☒ B. 2 [54%]
☐ C. 3 [12%]
☐ D. 4 [28%]

Correct

55% answered correctly

Explanation:



Proton **nuclear magnetic resonance** spectroscopy (^1H NMR) detects protons in a molecule when an external magnetic field and radio frequency are applied to a sample. The protons align their magnetic field either with (low-energy α spin state) or against (high-energy β spin state) the external magnetic field. Radio waves excite the protons from the α to the β spin state, and the peaks on the spectrum represent the energy difference between the spin states, known as the **effective magnetic field**.

Hydrogen atoms that are in **identical magnetic environments** within a molecule exhibit **rotational** or **planar** symmetry. These atoms are said to be **chemically equivalent** because they have the same atoms surrounding them, have the **same chemical shift**, and appear as a **single signal**. The **splitting pattern** of a signal (ie, how many peaks a signal is split into) is based on the $n + 1$ rule.

Compound **3** contains four aromatic hydrogen atoms (H_a – H_d). H_a and H_d are chemically equivalent because they are both adjacent to an aromatic C– NO_2 and an aromatic C–H, which is adjacent to an aromatic carbon with a side chain. Similarly, H_b and H_c are equivalent, being adjacent to identical chemical groups. Because there are two sets of equivalent aromatic hydrogen atoms and no other aromatic protons, there are two aromatic hydrogen signals (each appearing as a doublet) in the ^1H NMR spectrum.

(Choice A) There would be one signal in the ^1H NMR spectrum for the aromatic hydrogens in Compound **3** if the four aromatic protons were all equivalent to each other.

(Choice C) There are not three distinct (nonequivalent) aromatic hydrogen atoms in Compound **3**, therefore there cannot be three signals for aromatic protons in the ^1H NMR spectrum.

(Choice D) Although there are four aromatic protons in the structure of Compound **3**, some are chemically equivalent to each other, so fewer than four signals appear.

Educational objective:

Hydrogen atoms in a molecule that has the same types of atoms surrounding it and that exhibits rotational or planar symmetry are chemically equivalent and experience identical magnetic environments. Therefore, equivalent hydrogen atoms have the same chemical shift and appear as a single signal in the ^1H NMR spectrum.

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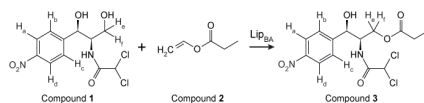


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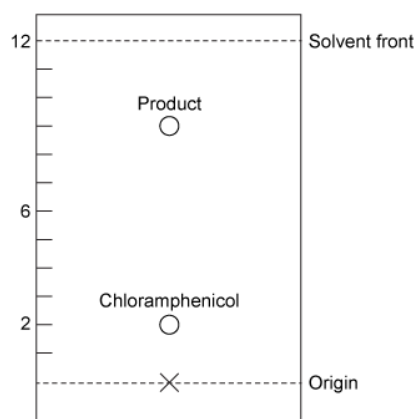


Figure 2 Thin-layer chromatography of the reaction mixture

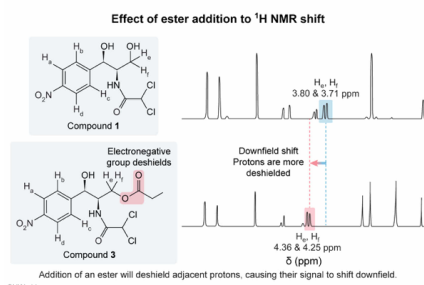
How does the addition of the ester to Compound **1** affect the shift of H_e and H_f? The signals for H_e and H_f will move:

- ✔ ☒ A. downfield because the ester deshields H_e and H_f. [56%]
- ☐ B. downfield because the ester shields H_e and H_f. [13%]
- ☐ C. upfield because the ester deshields H_e and H_f. [12%]
- ☐ D. upfield because the ester shields H_e and H_f. [17%]

Correct

56% answered correctly

Explanation:



Proton **nuclear magnetic resonance** spectroscopy (¹H NMR) applies an external magnetic field and radio waves to a sample and provides signals representing the protons in the sample. The location of signals in the ¹H NMR spectrum is known as the **chemical shift** and is given in parts per million (**ppm**) (typical range is 0–12 ppm).

The chemical shift depends on the electronic environment around the proton. An electron cloud around the protons **shields** them from the **external magnetic field**, causing an **upfield** signal (small chemical shift). An electronegative substituent withdraws electrons and **deshields** neighboring protons from the magnetic field, causing a **downfield** signal (larger chemical shift).

In Compound **1**, H_e and H_f are on a carbon adjacent to a hydroxyl group. Because oxygen is electronegative, it deshields the neighboring protons, causing a downfield shift (3.80 ppm and 3.71 ppm, respectively) from the chemical shift of alkyl protons (0–2 ppm). Transesterification of Compound **1** transforms the hydroxyl group into an ester (Compound **3**), which contains more electronegative atoms than the hydroxyl group. Therefore, the ester withdraws electrons and deshields H_e and H_f more so than the hydroxyl group, causing the signal to shift farther downfield (4.36 ppm and 4.25 ppm, respectively).

(Choice B) Although it is correct that the signals for H_e and H_f shift downfield, the ester is electronegative and therefore *deshields*, rather than shields, the protons.

(Choices C and D) Because an ester is electronegative, it withdraws electrons and deshields neighboring protons; therefore, the signals for H_e and H_f must shift downfield rather than upfield. A less electronegative atom would have to replace the oxygen adjacent to the carbon bonded to H_e and H_f for the signals to shift upfield.

Educational objective:

The location of signals in the ¹H NMR spectrum is known as the chemical shift, measured in ppm. The chemical shift depends on the electronic environment around protons. Protons with an electron cloud surrounding them are shielded from the magnetic field and exhibit an upfield signal whereas protons adjacent to electronegative substituents are deshielded from the magnetic field and have a downfield signal.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 402227

System:

Separation Techniques, Spectroscopy, and Analytical Methods

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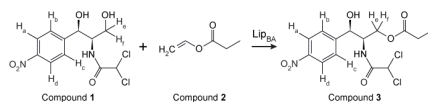


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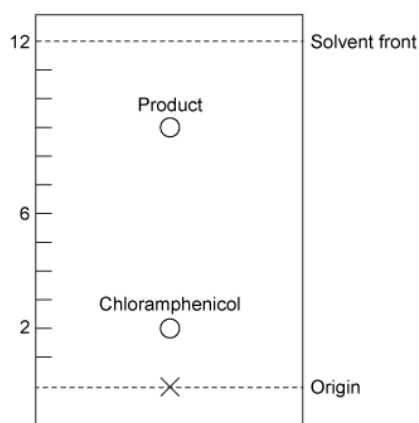


Figure 2 Thin-layer chromatography of the reaction mixture

Which of the following experimental conditions does NOT contribute to the highest conversion efficiency?

- ☐ A. Use of 1,4-dioxane as the solvent [2%]
☐ B. Reaction temperature of 55 °C [3%]
✓ ☒ C. Reaction time of 20 hours [76%]
☐ D. Enzyme concentration of 4.5 g/L [17%]

Correct

76% answered correctly

Explanation:

Transesterification conditions optimization

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Note: All trials were done with an enzyme concentration of 4.5 g/L. Trials where no enzyme was added did not yield any product and are not listed.

■ = variable that gives a highest conversion efficiency

■ = highest conversion efficiency

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Reaction condition optimization is important for achieving the efficient synthesis of a molecule. One variable should be evaluated at a time, and all others held constant so that only **one variable** is considered in **each trial**. The reaction conditions that give the highest conversion efficiency and purity are optimal.

Table 1 shows the results of the reaction condition optimization for the transesterification of chloramphenicol. The passage states that trials where enzyme was not added did not yield any reaction product whereas trials where 4.5 g/L of the enzyme Lip_{BA} was added produced the desired product (Compound **3**). Therefore, the enzyme is necessary for the transesterification reaction to occur and contributes to the high conversion efficiency (**Choice D**). The solvent 1,4-dioxane (**Choice A**) gave a considerably higher conversion efficiency (91% or higher) and purity than dichloromethane (72% conversion). Of the temperatures tested, 55 °C yielded the highest conversion efficiency (**Choice B**).

A reaction time of 10 hours, rather than 20 hours, gave the highest conversion efficiency at 55 °C (99%). A longer reaction time likely led to product degradation or side reactions. Therefore, a **20-hour reaction time** is **not part of the optimal set** of reaction conditions.

Educational objective:

When experimental conditions are optimized, one variable should be evaluated in each trial while other variables are kept constant. The conditions that give the highest conversion efficiency are optimal.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 402228

System:
Separation Techniques, Spectroscopy, and Analytical Methods

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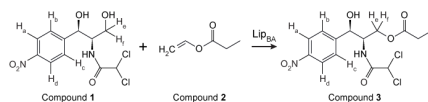


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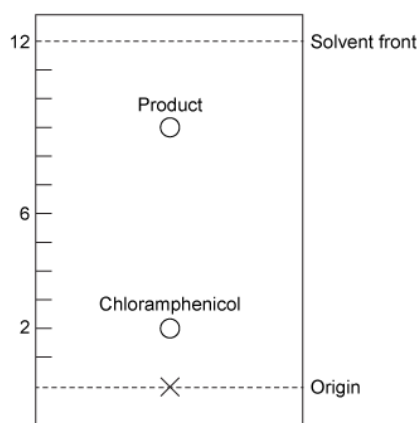


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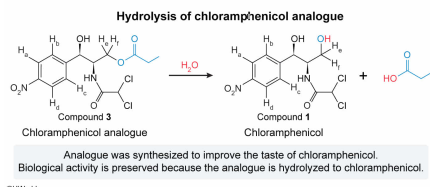
The major goal of synthesizing chloramphenicol analogues, such as Compound **3**, was most likely:

- ☐ A. to have greater biological activity than chloramphenicol. [1%]
- ✓ ☒ B. to improve the taste of chloramphenicol while preserving its biological activity. [93%]
- ☐ C. to avoid protection and deprotection steps in the synthesis. [1%]
- ☐ D. to create a compound that could be more easily purified than chloramphenicol. [3%]

Correct

94% answered correctly

Explanation:



Many drugs have a bitter taste and are unpleasant for patients to ingest. However, these bitter drugs can be "masked" as an analogue. One functional group can be modified to **improve the taste** of the drug, often without altering its effectiveness. In some cases, the analogue is converted back to the parent drug in the body. One such modification is **transesterification**, in which the alkoxy group (–OR group) on an ester and alcohol are switched. Esters are readily **hydrolyzed** back to the **alcohol** (along with a **carboxylic acid** byproduct) in the acidic environment of the stomach.

The passage states that chloramphenicol tastes bitter, making the drug unpleasant for patients. To reduce the bitterness, an analogue of chloramphenicol was synthesized via transesterification, as shown in Figure 1. Because chloramphenicol analogues with a substituent on the **primary hydroxyl group**, such as Compound **3**, are hydrolyzed back to the parent drug *in vivo*, taking an analogue has the same effect as taking chloramphenicol. Therefore, the **biological activity** of chloramphenicol is **preserved**.

(Choice A) The passage states that the analogue is hydrolyzed *in vivo* to chloramphenicol. Therefore, chloramphenicol, not the analogue, is responsible for the biological activity.

(Choice C) Avoiding protection and deprotection steps, which can lengthen a synthesis significantly, is accomplished using the enzyme Lip_{BA} lipase in the synthesis. However, avoiding these steps was not the goal of synthesizing the analogues.

(Choice D) Although ease of purification is an important factor to consider when designing the synthesis of a molecule, the results described in the passage do not give any indication that Compound **3** was easier or more difficult to purify than Compound **1**.

Educational objective:

Transesterification is a reaction in which the alkoxy group (–OR group) on an ester and alcohol are switched. The ester can be hydrolyzed back to the alcohol (along with a carboxylic acid byproduct) in acidic conditions.

Time spent: 0 sec
Subject:
Organic Chemistry

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System:
Separation Techniques, Spectroscopy, and Analytical Methods